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Original Article

San Huang Xiao Yan recipe modulates the HMGB1-mediated abnormal inflammatory microenvironment and ameliorates diabetic foot by activating the AMPK/Nrf2 signalling pathway

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A R T I C L E I N F O

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A B S T R A C T

Background: Diabetic foot (DF) is one of the serious complications of diabetes and lacks of therapeutic drugs. Abnormal and chronic inflammation promoting foot infection and wound healing delay are the main pathogenesis of DF. The traditional prescription San Huang Xiao Yan Recipe (SHXY) has been used in the clinical treatment of DF for several decades as approved hospital experience prescription and showed remarkable therapeutic effect, but the mechanisms by which SHXY treats DF are still unclear.

Purpose: Objectives of this study were to investigate SHXY anti-inflammatory effect on DF and explore the molecular mechanism for SHXY.

Methods: We detected the effects of SHXY on DF in C57 mouse and SD rat DF models. Animal blood glucose, weight and wound area were detected every week. Serum inflammatory factors were detected by ELISA. H&E and Masson's trichrome were used to observe tissue pathology. Single-cell sequencing data reanalysis revealed the role of M1 macrophages in DF. Venn analysis showed the co-target genes between DF M1 macrophages and compound-disease network pharmacology. Western blotting was used to explored target protein expression. Meanwhile, RAW264.7 cells were treated with drug-containing serum of SHXY to further unravel the roles of target proteins during high glucose-induced inflammation *in vitro*. The Nrf2 inhibitor ML385 was used on RAW 264.7 cells to further explore the relationship between Nrf2, AMPK and HMGB1. The main components of SHXY were analysed by HPLC. Finally, the treatment effect of SHXY on DF were detected on rat DF model.

Results: In vivo, SHXY can ameliorate inflammatory, accelerate wound healing and upregulate expression of Nrf2, AMPK and downregulate of HMGB1. Bioinformatic analysis showed that M1 macrophages were the main inflammatory cell population in DF. Moreover, the Nrf2 downstream proteins HO-1 and HMGB1 were potential DF therapeutic targets for SHXY. *In vitro*, we also found that SHXY increased AMPK and Nrf2 protein levels and downregulated HMGB1 expression in RAW264.7 cells. Inhibiting the expression of Nrf2 impaired the inhibition effect of SHXY on HMGB1. SHXY promoted Nrf2 translocation into the nucleus and increased the phosphorylation of Nrf2. SHXY also inhibited HMGB1 extracelluar release under high glucose. In rat DF models, SHXY also exhibited significant anti-inflammatory effect.

Abbreviations: AMPK, AMP-activated protein kinase; CRP, C-reactive protein; DF, diabetic foot; FGF, fibroblast growth factor; HG, high glucose; HMGB1, high mobility group box-1 protein; HO-1, hemoxygenase-1; HPLC, high-performance liquid chromatography; IL-6, interleukin (IL)-6; Met, metformin; Nrf2, nuclear factor E2-related factor 2; p-AMPK, Phosphorylated-AMP-activated protein kinase; SA, Staphylococcus aureus; SHXY, San Huang Xiao Yan Recipe; STZ, streptozotocin; TNF-α, tumor necrosis factor α.

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Conclusion: The SHXY activated AMPK/Nrf2 pathway to suppress abnormal inflammation on DF via inhibiting HMGB1 expression. These findings provide novel insight into the mechanisms by which SHXY treats DF.

Introduction

Diabetic foot (DF) is a serious complication of diabetes that lacking therapeutic drugs. Persistent hyperglycaemia promotes abnormal inflammation aggravating foot infection and wound healing delay in DF (Bader, 2008; Baltzis et al., 2014). Globally, it is estimated that a diabetic amputation occurs every 20 s. The annual mortality rate for diabetic foot ulcer patients is as high as 11% and 22% for amputees (Gao et al., 2020). Unfortunately, in the past few decades, little improvement has been made in preventing the morbidity and disability of DF. Therefore, the more effective and specific drugs are urgently needed for the treatment of DF.

Many studies have indicated that Chinese herbal formulas display excellent curative effects in treating chronic diseases including DF. Professor Jiuyi Xi, the founder of the Department of Vascular Diseases of Shanghai TCM-Integrated Hospital and the Institute of Vascular Diseases of Integrated Traditional Chinese and Western medicine. He invented the hospital experience prescription San Huang Xiao Yan Recipe (SHXY), which consists of 11 herbs process a good curative effect on DF. Clinical study has shown that SHXY can significantly improve abnormal inflammation in DF patients by reducing the levels of inflammatory factors in the serum (J. and Q. 2018). Moreover, the combined use of SHXY and topical medications applied to the diabetic foot ulcer can also achieve the effect of promoting wound healing (Leng, 2021). Based on significant curative effects of SHXY on treating DF, its therapeutic mechanism deserved to be further explored.

Nuclear factor erythroid 2 related Factor 2 (Nrf2) is a nuclear transcription protein. Under normal physiological conditions, Nrf2 combined with Keap1 stably exists in the cytoplasm of cells. Oxidative stress promotes Nrf2 translocation into the nucleus and binding to ARE elements, further promoting the expression of downstream antioxidant genes such as HO-1, NQO1 and SOD, which play antioxidant role (Bellezza et al., 2018). In addition, Nrf2 plays an important role in regulating innate immunity by suppressing the expression of proinflammatory genes and enhancing the conduction of anti-inflammatory signals (Kim et al., 2010). At the same time, recent studies have found that overexpression of Nrf2 or Nrf2 activator treatment for DF rats can improve the inflammatory response, promote the expression of growth factors in wound tissue, and accelerate wound healing (Li et al., 2019, 2018b).

Adenosine 5′-monophosphate (AMP)-activated protein kinase (AMPK), an AMP-dependent protein kinase, is a key molecule in the regulation of bioenergy metabolism and the core of the study of diabetes and other metabolism-related diseases. Previous studies have shown that AMPK activator metformin can accelerate DF wound repair by reducing tissue inflammation and promoting epithelial cell regeneration (Qing et al., 2019). Many studies have suggested AMPK could be an upstream protein of Nrf2. In the pathogenesis of diabetic cardiomyopathy, sulforaphane inhibits cardiomyocytes ferroptosis by activating AMPK/Nrf2 pathway (Wang et al., 2022). Dioscin can activate AMPK/Nrf2 to inhibit NLRP3/GSDMD-induced mMEC pyroptosis (Ran et al., 2020). Metformin combined with exercise training significantly alleviated hepatic lipid accumulation through upregulating the AMPK-Nrf2-HO-1 signaling pathway in *db/db* mice (Zhang et al., 2022).

High mobility group box-1 protein (HMGB1) not only plays an important role in the assembly, transcription and repair of genes in the nucleus, but also acts as an important late inflammatory mediator when it is actively secreted by cells or released into the extracellular space by damaged and necrotic cells. It binds to membrane receptors such as the receptor for advanced glycation end products (RAGE) and exerts its biological effects in infection, inflammation and the immune response (Chen et al., 2004). In diabetic nephropathy, HMGB1 further promotes the release of inflammatory factors such as IL-6, TNF-α and IL-1β by activating the NF-κB signalling pathway under lipopolysaccharide stimulation (Chen et al., 2015). In the study by Yasser et al., HMGB1 was found to be highly expressed by PCR in the foot tissues of 60 DF patients (Hafez et al., 2018).

In previous work, we also analysed the relationship between SHXY and DF by network pharmacology. We separately found 1969 drug targets of SHXY and 670 disease targets of DF. A total of 238 overlapping targets were detected by Venn (http://bioinfogp.cnb.csic.es/tools/venny/index.html). These targets were further analysed by Metascape (https://metascape.org/gp/index.html#/main/step1). Preliminary analysis suggested that suggested the genes of Nrf2, HO-1, and HMGB1 were involved in ARE-RAGE pathway (Fig. S1).

Based on the above literature research and network pharmacology analysis, we preliminarily speculated that SHXY could treat DF by activating the AMPK/Nrf2/HMGB1 pathway. In the present research, we separately constructed two animal models induced with streptozotocin (STZ) simulating the clinicopathological status of DF and treated these models with SHXY separately to explored the inflammatory microenvironment regulation effect of SHXY in DF. Single-cell sequencing data reanalysis and network pharmacology were used to explained the influence of macrophage in DF. High glucose-induced diabetic inflammation models of RAW264.7 cell *in vitro* were also used to verify our speculation. Meanwhile expressions of AMPK, Nrf2 and HMGB1 also were detected *in vitro* and *in vivo* experiments.

Method and materials

Ethics statement

All of the animal procedures involving mice, such as housing and care, and experimental protocols were approved by the Tongji University Medical College Northern Animal Care Committee and Use Committee. All procedures performed on the rats and mice were conducted according to the guidelines from the National Institutes of Health (No. TJ-HB-LAC-2022–15 and No. TJ-HB-LAC-2022–16).

Composition of the SHXY recipe

SHXY recipe (Z05050568, 7 g per pack, two times daily) was composed of 11 herbs (Table 1) and produced by Shanghai Liantang Pharmaceutical.

Rat diabetic foot infection models

SD rats were purchased from Shanghai Sipur-Bikai Laboratory Animal Co., Ltd, China, and fed in the SPF Laboratory Experimental Animal Center of Tongji University Medical College, Shanghai, China. All rats stayed in a specific pathogen-free environment with a constant to temperature (25 ± 3 °C), humidity (55 ± 10%) and light cycle (12:12 h light-dark), and were fed a purified diet and water. Then 24 rats were randomly assigned to the control group, STZ+SA group and SHXY group ($n = 8$ per group). The STZ+SA group was established through intraperitoneal injection with 45 mg/kg STZ (S0130–1 G, Sigma, USA) (Tuchscherr et al., 2018), while the control group received the same volume of saline instead. After 3 days, the blood glucose of the rats was detected by a blood glucose measuring instrument. Blood glucose levels were maintained between 16.7 and 33.3 mmol/l for one week, which suggested that diabetes model was successfully established. The STZ+SA

Table 1
Information for SHXY granules composition.

Chinese name	Herb English name	Herb Latin name
DA HUANG (DH)	Root and rhziome of Sorrel Rhubarb	Radix et Rhizoma Rhei
HUANG QIN (HQ)	Root of Baikal skullcap	Radix Scutellariae
HUANG BAI (HB)	Amur Corktree Equivalent plant: Phellodendron chin	Phellodendron amurense
BAN LAN GEN (BLG)	Radix Isatidis seu Baphicacanthi	Radix Isatidis seu Baphicacanthi
LIAN QIAO (LQ)	Fructus Forsythiae	Weeping Forsythia Capsule
QUAN SHEN (QS)	Rhizome of Bistort	Rhizoma Bistortae
YA ZHI CAO (YZC)	All-grass of Common Dayflower	Herba Commelie
GAN CAO (GC)	Root of Ural Licorice	Radix Glycyrrhizae
HUANG LIAN (HL)	Rhizome of Chinese Goldthread	Rhizoma Coptidis
SHAN ZHA (SZ)	Chinese Hawthorn Equivalent plant: Crataegus pinna	Crataegus pinnatifida
HU ZHANG (HZ)	Rhizome of Gaint Knotweed	Rhizoma Polygoni cuspidati

group was injected with 1×10^8 CFU ATCC25923 Staphylococcus aureus (SA) into the mouse footpads under isoflurane anaesthesia (Tuchscherr et al., 2018). On Day 2 after SA injection, these rats were divided into three groups, one of the groups was gavaged with SHXY 1.47 g/kg for 4 weeks and other one was gavaged with the equal volume normal saline for the same time. On Day 35, all rats were anaesthetized with 10% chloral hydrate intraperitoneally, and foot tissues were collected for subsequent experiments.

Mouse diabetic wound models

The mouse diabetic ulcer model was establishment according to previous researches (Fadini et al., 2016). C57 mice were purchased from Hangzhou Medical College and fed in the SPF Laboratory Experimental Animal Center of Tongji University Medical College, Shanghai, China. All mice stayed in a specific pathogen-free environment with temperature (25 ± 3 °C), humidity (55 ± 10%) and light cycle (12:12 h light-dark), and were fed a purified diet and water. Then, 25 mice were randomly assigned to the normal wound group (NW), diabetic wound group (DW), SHXY group (SHXY), metformin group (MET) and FGF group (FGF). All groups were established through intraperitoneal injection with 100 mg/kg streptozotocin (S0130–1 G, Sigma, USA) for three days except the NW group. After 3 days, the blood glucose of mice was detected by a blood glucose measuring instrument. Blood glucose levels were maintained between 16.7 and 33.3 mmol/l for one week, which suggested that the diabetes model was successfully established. Then, all mice were used for open wound induction. On the day of open wound induction (Day 0), each mouse was anaesthetized with 2% isoflurane. The surface skin of the back was shaved and cleaned with a 70% ethanol wipe. A diameter1 7 mm x 8 mm rectangular full thickness wound was created in the skin of the back of each mouse using a scalpel. On Day 2 after diabetic wound model established, mice were divided into five groups, normal wound group (NW), diabetic wound group (DW), diabetic wound group was gavaged with San Huang Xiao Yan recipe (SHXY) (2.12 g/kg, Z05050568, Shanghai Liantang Pharmaceutical, China), diabetic wound group was gavaged with metformin (Met) (200 mg/kg, 214–230–6, Noble Biology, China) and diabetic wound group was applied externally to the wound with fibroblast growth factor (FGF) (S10980077, Essex, China) for 2 weeks. On Day 7 and Day 14 mice were anaesthetized with 10% chloral hydrate intraperitoneally, and back skin tissues and serum were collected for subsequent experiments.

Wound healing assessment

The size of the wound of mice was observed on Day 0, Day 7, Day 14 and imaged by a digital camera at specified time points in a 2-week experimental period because there were a few wounds showed a tendency to completely heal within two weeks, but no wounds completely healed. Afterwards, the area of the wound was quantitated by ImageJ software and calculated as follows: wound area = (actual wound area/original wound area) × 100%.

Histological analysis

The mouse wound skins were collected within 6–8 mm from the wound centre for paraffin embedding and protein extraction. The regenerated skins were fixed in 4% paraformaldehyde for 12 h, embedded in paraffin, sliced to 5 mm thickness and then dewaxed for histological analysis. ImageJ software was used to analyse H&E staining of the epidermal structure and inflammatory response of the wound healing layer. Inflammation score of 1 to 10 is based on the infiltration of inflammatory cells.

Masson and IHC

Using 10% formaldehyde solutions fixed mouse ulcer skin tissues were fixed for 24 h, and then skin tissues were embedded in paraffin. The paraffin blocks were cut at 5 μm using a microtome. Then, skin tissue slices were stained with Masson trichrome staining and H&E staining. Using IHC analysis was used to detect the expression of Nrf2, AMPK, p-AMPK and HMGB1. Biotin-Streptavidin HRP Detection Systems (SPN-9001, ZSGB-BIO, China) were used for this experiment and steps according to the manufacturer's instructions. Tissue sections were incubated with primary anti-Nrf2 (AF0639, Affinitiy, China), anti-AMPK (AF6423, Affinitiy, China), anti-p-AMPK (AF3423, Affinitiy, China) and anti-HMGB1 (10,829–1-AP, Proteintech, USA) antibodies at corresponding appropriate ratios of 1:200, 1:200, 1:200 and 1:200 and kept at 4 °C overnight in a humidifying box. The next day, the sections were incubated with the general secondary antibody HRP Goat Anti-Rabbit IgG ($H + l$) (AS014, ABclonal, China) were added to slices for 1 h at room temperature, and diaminobenzidine (P0202, Beyotime, China) was added as a visualizing agent. Nuclear staining was performed with hematoxylin, and a light microscope (DMI8, Leica, Germany) was used to detect target proteins. Collagen volume fraction (CVF) and average Optical Density (AOD) were quantified with Image J.

Single-cell analysis of M1 macrophages

scRNA-Seq data were obtained from NCBI's Gene Expression Omnibus (GEO) and the GEO accession number was GSE165816. Gene Ontology (GO) analysis was conducted by the R package clusterProfiler (version: 4.3.2.991) with default parameters, as well as the Gene Set Enrichment Analysis (GSEA).

SHXY-containing serum preparation

SHXY-containing blood was collected from the SHXY rat group (SHXY ($n = 5$): 2.94 g/kg/d, double the last dose) group after dosing at 0.5, 1, and 2 h after the last dose. Then the blood was centrifuged at 4 °C (12,000 rpm-10 min) to obtain the upper layer of serum. Serum was separated pack and stored at −80 °C for cell experiments.

Cell line and culture

In vitro, RAW264.7 cells (ATCC, USA) were grown in 1640 medium (R8758, Sigma, USA) supplemented with 10% foetal bovine serum (FBS; F8687, Sigma, USA), and the cells were incubated at 37 °C with 5% CO_2. High glucose group (HG) was incubated with 30 mM high glucose 1640

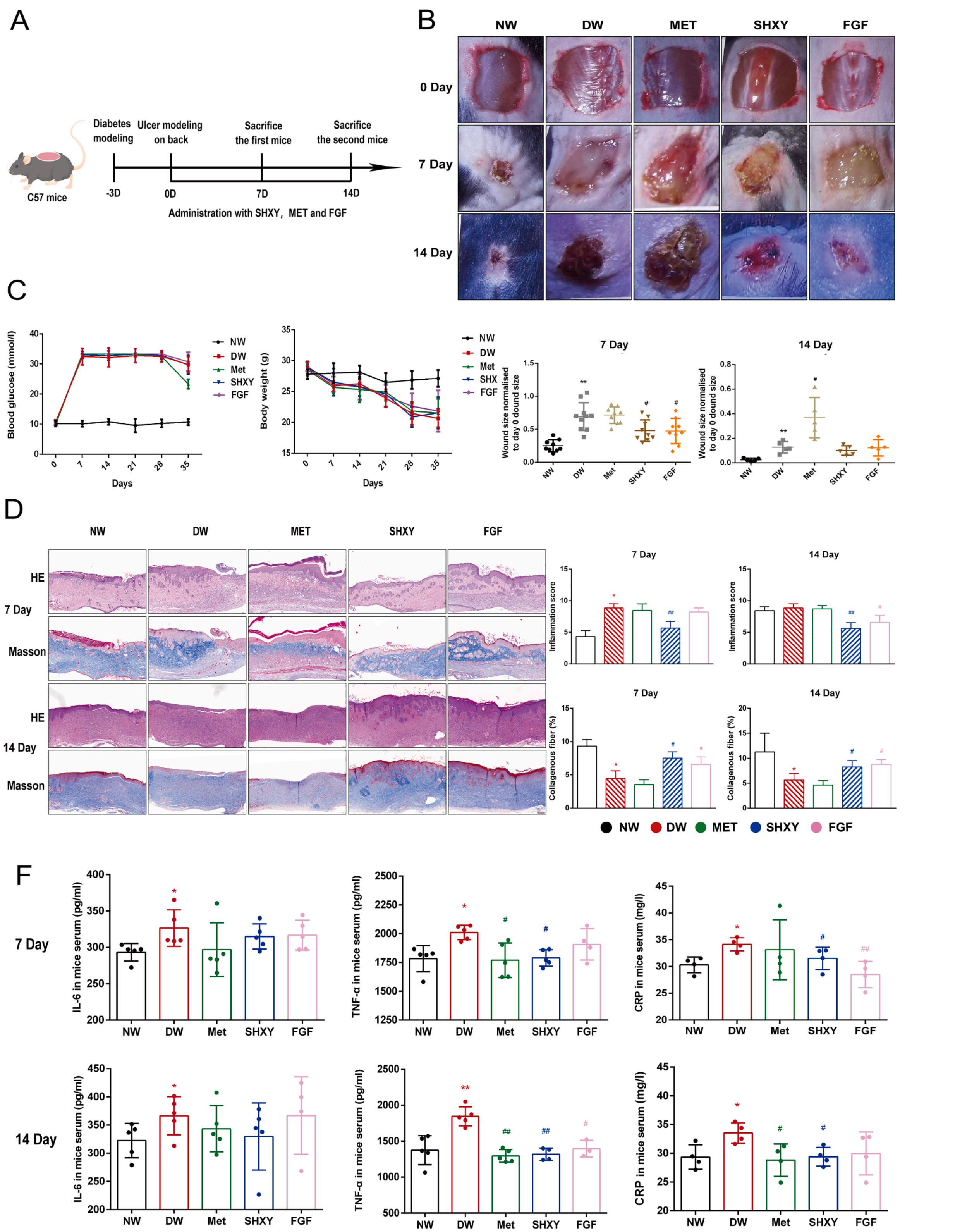

Fig. 1. Administration of SHXY accelerated wound healing and alleviated inflammation in the DF mouse wound model. (A) After the DF mouse wound model was established by STZ and puncher. Mice were intragastrically administered by SHXY (2.12 g/kg), metformin (0.2 g/kg) and FGF (spray every other day) for 14 days. Mice were respectively euthanized on Day 7 and Day 14. (B) Wound healing recorded by digital camera from each group. Quantitative statistics of wound size. (C) Blood glucose of mice in different groups, $n = 5$ per group. Body weight of rats in different groups, $n = 5$ per group. (D) H&E and Masson staining of wound skin from each group, respectively. Scale bar, 200 μm. Quantitative Statistics for H&E and Masson staining, $n = 3$ per group. (E) Inflammatory cytokine levels in serum, $n \geq 3$ per group. Values are shown as the mean ± SD of 3 mice. * $p < 0.05$, ** $p < 0.01$ *vs* NW group and # $p < 0.05$, ## $p < 0.01$ *vs* DW group.

medium (Meilin et al., 2010) for 24 h, SHXY group (SHXY) was treated with 10% SHXY-containing serum for other 24 h after stimulated with 30 mM high glucose 1640 medium for 24 h, metformin group (Met) was treated with 2 μM metformin (Qing et al., 2019) (214–230–6, Noble Biology, China) for other 24 h after stimulated with 30 mM high glucose 1640 medium for 24 h and FGF group (FGF) was treated with 100 ng/ml FGF (S10980077, Essex, China) (Yu et al., 2016) for other 24 h after stimulated with 30 mM high glucose 1640 medium for 24 h The above method of administration of RAW 264.7 cells were used for reactive oxygen species (ROS) detection, *in vitro* western blot detection, Immunofluorescence analysis and cell Cytokine analysis.

Reactive oxygen species (ROS) detection

Cells were incubated with chloro-methyl-dichlorofluorescein diacetate (CM-DCFDA, 10 mM) (Molecular Probes, Eugene, OR, USA) for 30 min at 37 °C. Samples were then centrifuged at 1000 × g for 3 min, and the pellets were resuspended in 500 μl phosphate-buffered saline (PBS). Measurements and analysis were performed on a FACS Calibur (Beckman, San Jose, USA) flow cytometer. ROS immunofluorescence was conducted after flow cytometer detection.

Western blot analysis (WB)

Rat foot muscle tissue, mouse skin tissues and cultured cells were homogenized in lysis buffer (P0013K, Beyotime, Shanghai, China) containing proteinase inhibitors (PMSF; P0100, Solarbio, China). The protein concentration in the supernatant was measured using a BCA protein assay kit (P0100, Beyotime, China). All the protein samples were electrophoresed on 10% SDS-PAGE gels and transferred onto PVDF membranes (IPVH00010; Millipore, USA). The blots were incubated with the following primary antibodies: Nrf2 (1:1000; ab92946, Abcam) & (1:1000; AF0639, Affinity), AMPK (1:1000; ab92946, Abcam), p-AMPK (1:1000; ab133448, Abcam), p-Nrf2 (1:1000; 12721S, Affinity), HMGB1 (1:1000; 10,829–1-AP, Proteintech), HO-1(1:1000; AF5393, Affinity), Ser40 p-Nrf2 (1:1000; DF7519, Affinity), H3 (1:1000; AF0863, Affinity), tublin (1:1000; AF7011, Affinity), and anti-β-actin (1:1000; AP0060, Bioworld) antibodies overnight at 4 °C. The next day, all blots were incubated with goat anti-mouse IgG (*H* + l*)* (AS003, Abclonal, USA) or goat anti-rabbit IgG (*H* + l*)* (AS014, Abclonal, USA) for 1 h at room temperature after washing five times (each time for 5 min). The target proteins with similar molecular weight were eluted with the antibody stripping solution (P0025–250 ml, Beyotime, China), and then re-incubated with the other primary antibody overnight. The blots were visualized by a protein imager (UVPChem Studio touch, Analytik Jena, Germany) and quantified with the Adobe photoshop CS5 and GraphPad Prism 6.02.

Cytokine analysis

Under general anaesthesia, 5000 ul of blood was extracted from each rat. Serum samples were then centrifuged at 3000 rpm for 10 min. The serum was collected and diluted at 1:20. Twenty microlitres of each serum sample was used to determine the IL-6, TNF-α and CRP levels according to the instructions of the ELISA Kit (70-EK306/3–96, 70-EK382/3–96 and 70-EK394–96, Multisciences, China), and the corresponding OD values were detected by a microplate reader (Spectra Max ID3, MoleCular Devices, USA). Under general anaesthesia, 500 μl of blood was extracted from each mouse. Serum samples were then centrifuged at 3000 rpm for 10 min. The serum was collected and diluted at 1:20. Twenty microlitres of each serum and cellular supernatant sample was used to determine the HMGB1, IL-6, TNF-α and CRP levels according to the instructions of ELISA kits (BPE20487, YX-091,206 M, YX-201,407 M and YX031816, Liangdun and Yanjin Biological, China), and the corresponding OD values were detected by a microplate reader (Spectra Max ID3, MoleCular Devices, USA).

Immunofluorescence analysis (IF)

Cells were cultured for three days, the possess was the same with, washed three times with PBS at room temperature (10 s each time), then fixed with 4% formaldehyde for 20 min and permeated with 0.1% Triton X-100 (Sigma-Aldrich, USA) for 20 min. After blocking with 5% BSA for 30 min at room temperature, primary antibodies Nrf2 (1:300; AF0639, Affinity) and HMGB1(1:300; sc-56,698, Santa) were incubated with the cells at 4 °C overnight. Then, the cells were incubated with FITC-conjugated goat anti-rabbit IgG (ZF-0311; Beijing, China, 1:200) and Cy3-conjugated goat anti-mouse IgG (BL10761; Beijing, China, 1:200) for 1 h at 37 °C keeping in darkness. Cells were washed three times with PBS (10 s each time), followed by incubation with 6-diamidino-2-phenylindole (DAPI) (C1005; Beyotime, Shanghai, China) for 5 min to stain the nucleus. Cells were visualized with a Leica immunofluorescence microscope (Leica, STELLARIS, Germany) and identical acquisition parameters at 10 μm.

High performance liquid chromatography (HPLC) analysis of SHXY

Rhubarb (21,040,411), *Scutellaria baicalensis* (20,121,111), golden cypress (21,050,141), *Radix isatidis* (21,020,621), *Fructus forsythiae* (20,210,541), *Polygonum bistorta* (19,032,381), dayflower (20,081,611), liquorice (20,120,641), *Coptis chinensis* (21,010,861), hawthorn (21,020,481) and *Polygonum cuspidatum* (20,120,521), were purchased from Jiangyin Tianjiang Pharmaceutical for Food and Drug Control (Shanghai, China). An Acquity UPLC® BEH C18 column (100 mm × 2.1 mm, 1.7 μm) was used with a Dionex Ultimate 3000 high-pressure liquid system. The mobile phase consisted of 0.1% formic acid water and methanol, and the elution procedure was 0~4 min, 4% methanol. 4~12% methanol for 4~10 min; 10~30 min, 12%~70% methanol; 30~35 min, 70% methanol; 35–38 min, 70%−95% methanol; 38–42 min, 95% methanol; 42~45 min, 4% methanol. The flow velocity was 0.3 mLmin-1. The column temperature was 40 °C, and the sample size was 2 μl. A Q Exactive quadrupole-electrostatic field orbit-trap high resolution mass spectrometer was used as an instrument for analysis and detection. The ion source was an electrospray ion source (H-ESI), the auxiliary gas flow rate is 13 arb, auxiliary heater temperature was 300 °C, ion transfer tube temperature was 320 °C, automatic gain control (AGC) was 106, and the s-lens RF level was 50. Positive ion mode: sheath gas (N2) flow rate 35 arb, spray voltage 3.5 kV; Negative ion mode: sheath gas (N2) flow rate 35 arb, spray voltage 2.5 kV. The scanning mode was full MS/SIM mode with positive and negative ions, the resolution was 70,000 FWHM, and the scanning range of both positive and negative ions was *m/z* 80~1200.

Statistical analysis

All data are expressed as the mean ± standard deviation (SD), using the social science statistical software package for data analysis. The two groups of data were compared with the independent sample t-test and comparison of multiple sets of data were compared using a one-way analysis of variance analysis. The column graphs were drawn using GraphPad Prism 6 software. $p < 0.01$ or $p < 0.05$ was considered statistically significant.

Results

SHXY ameliorated inflammation in DF mice

A DF wound mice model was established to evaluate the effect of SHXY for two weeks (Fig. 1A). In this model, SHXY had no significant influence on improved HG(Fig. 1C). Wound healing assessment preliminarily suggested that SHXY promoted wound healing compared with DW group (Fig. 1B). Skin histological studies showed that SHYX reduced inflammatory cell infiltration and promoted ECM formation in wound

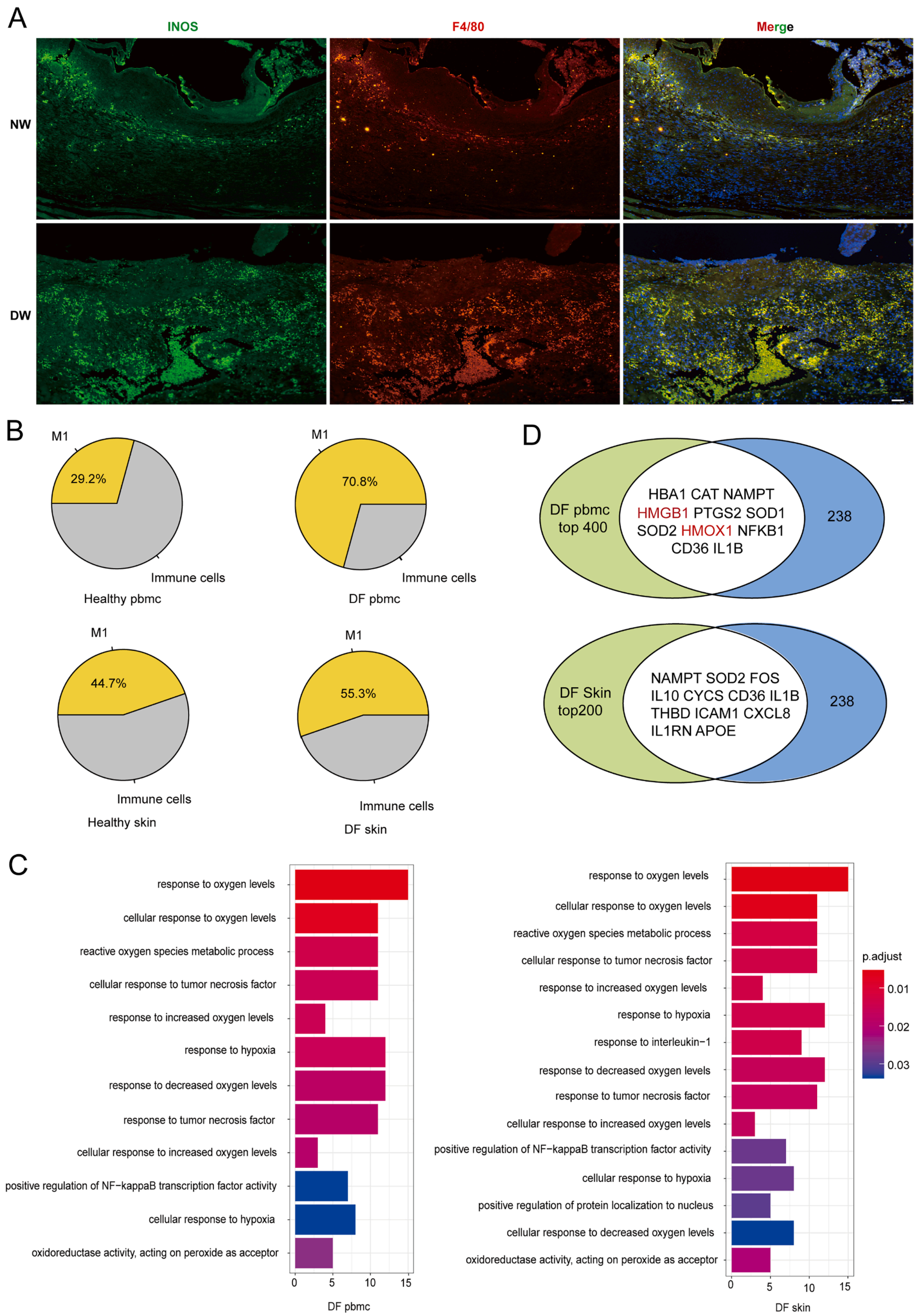

Fig. 2. Single-cell sequencing analysis of M1 macrophages in DF PBMCs and skin. (A) Immunofluorescent staining for the M1 macrophage biomarkers F4/80 and iNOS. Scale bar, 50 μm. (B) M1 macrophage proportion analysis in PBMCs and skin immune cells. (C) GO analysis for M1 macrophages in DF patients PBMCs and skin. (D) Venn analysis of network pharmacology targets and single-cell targets of M1 macrophages.

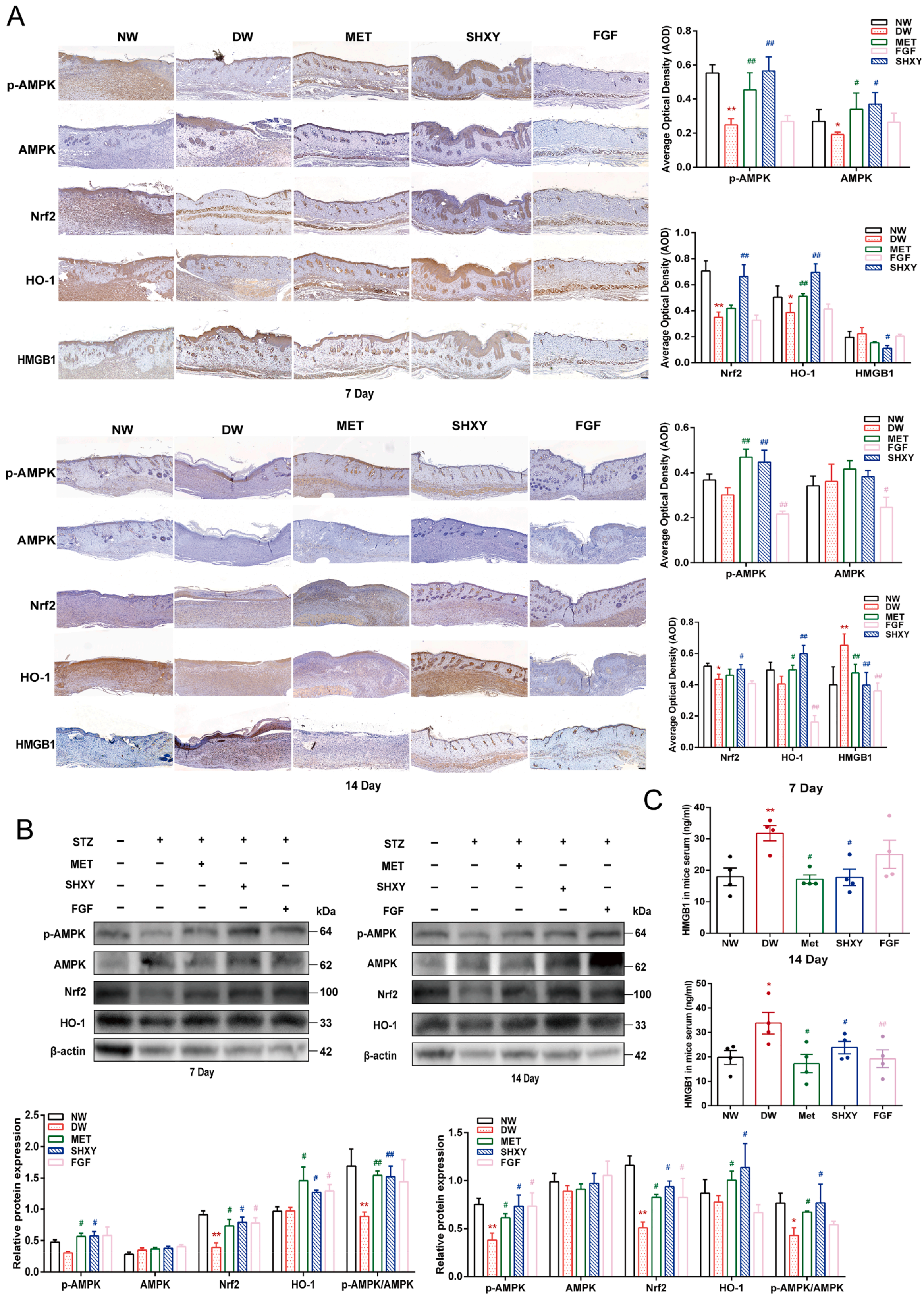

Fig. 3. Administration of SHXY upregulated the protein expression of p-AMPK, Nrf2, HO-1 and downregulated HMGB1 expression in the DF mouse wound model. (A) Relative protein expression levels of p-AMPK, AMPK, Nrf2, HO-1 and HMGB1 on Day 7 and Day 14 were separately measured by IHC. Quantitative statistics of IHC. (B) Relative protein expression levels of p-AMPK, AMPK, Nrf2, HO-1 and HMGB1 on Day 7 and Day 14 were separately were measured by WB. Quantitative statistics of WB, $n = 3$ per group. (C) Serum HMGB1 detected by ELISA kit. Values are shown as the mean ± SD of 3 mice. * $p < 0.05$, ** $p < 0.01$ *vs* NW group and $^{\#}$ $p < 0.05$, $^{\#\#}$ $p < 0.01$ *vs* DW group.

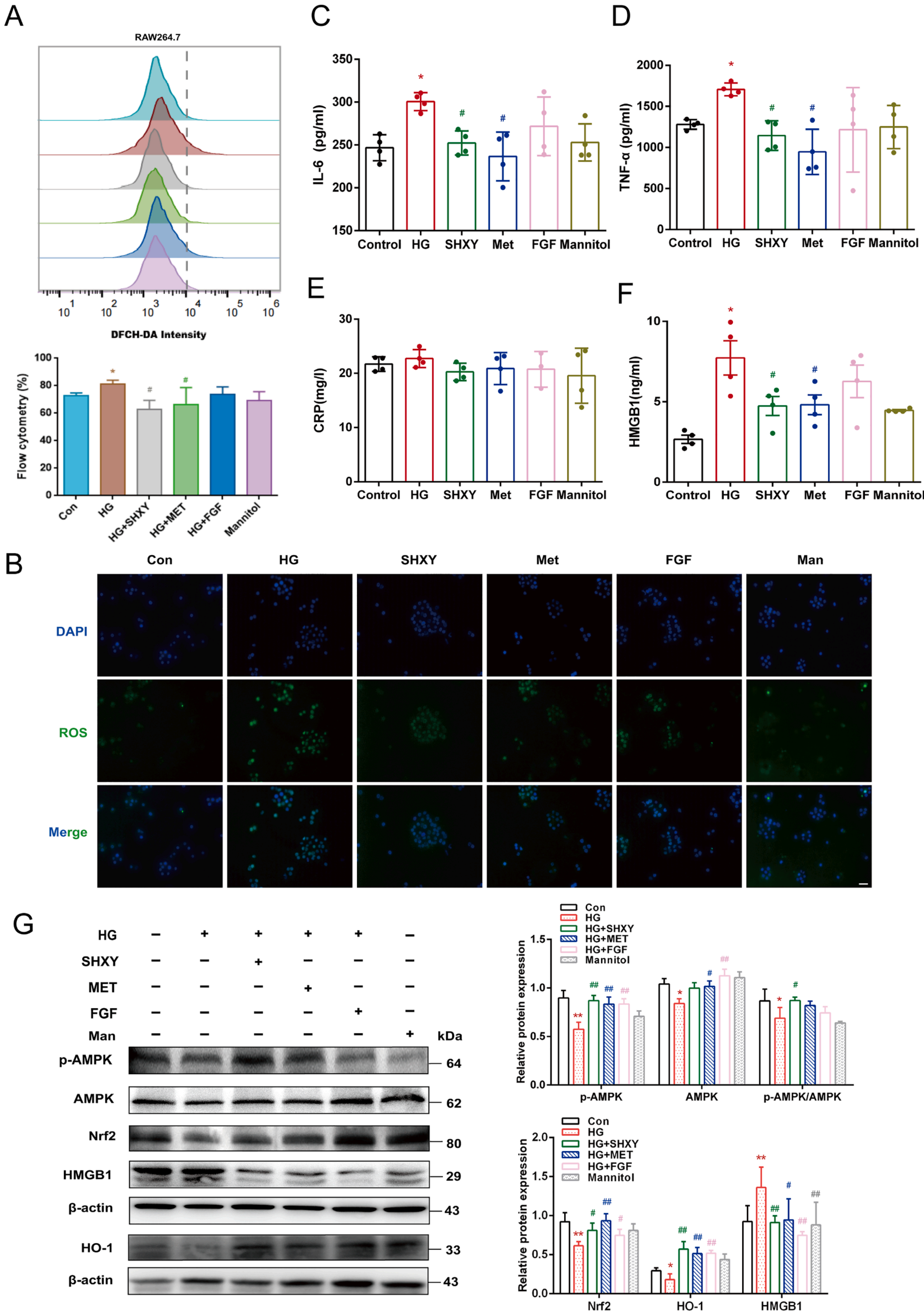

Fig. 4. SHXY drug serum upregulated the protein expression of p-AMPK, Nrf2, and HO-1 and inhibited HMGB1 in HG-stimulated RAW264.7 cells. (A) ROS detection by flow cytometry. (B) ROS detection by Immunofluorescence. Scale bar, 50 μm. (C) The IL-6, TNF-α, CRP and HMGB1 protein levels in cellular supernatant by ELISA kit, $n = 4$. (D) Relative protein expression levels of p-AMPK, AMPK, Nrf2, HO-1 and HMGB1 during HG stimulation of RAW264.7 cells were measured by WB. Quantitative statistics of WB, $n = 3$. Values are shown as the mean ± SD, $p < 0.05$, ** $p < 0.01$ *vs* control group and $^{\#} p < 0.05$, $^{\#\#} p < 0.01$ *vs* HG group.

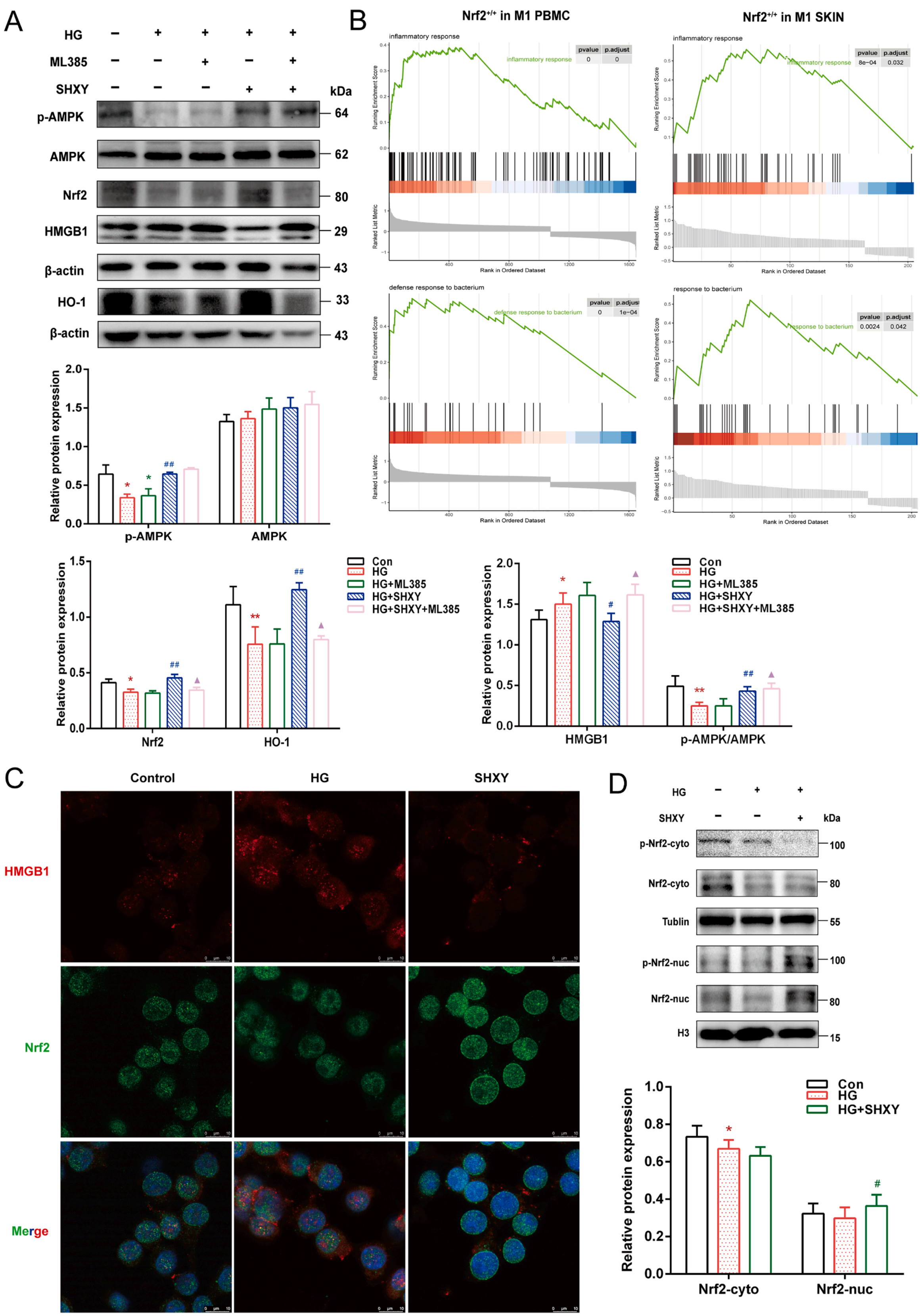

Fig. 5. SHXY drug serum promotes Nrf2 translocation from the cytoplasm to the nucleus and inhibits HMGB1 extracellular release. (A) Relative protein expressions levels of p-AMPK, AMPK, Nrf2, HO-1 and HMGB1 during HG stimulation of RAW264.7 were measured by WB. Quantitative statistics of WB, $n = 3$. (B) GSEA analysis for Nrf2 in M1 macrophages (C) Immunofluorescent staining for Nrf2 (Green dyeing), HMGB1 (Red dyeing) and cell nucleus (Blue dyeing) location detection during SHXY treating RAW264.7. Scale bar, 10 μm. (D) Relative protein expression levels of p-Nrf2 and Nrf2 in cytoplasm and nucleus during SHXY treating RAW264.7 cells were measured by WB. Quantitative statistics of WB, $n = 3$. Values are shown as the mean ± SD. * $p < 0.05$, ** $p < 0.01$ *vs* control group, $^{\#}$ $p < 0.05$, $^{\#\#}$ $p < 0.01$ *vs* HG group and Δ $p < 0.05$, ΔΔ $p < 0.01$ *vs* SHXY group.

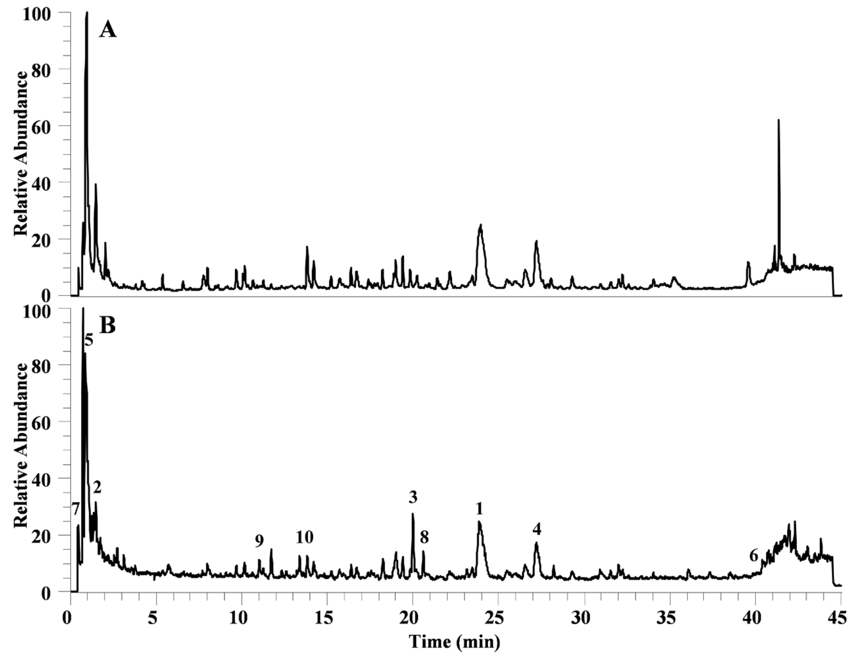

Fig. 6. Total ion chromatograms of SHXY prescript in negative. (A) and postive (B) ion modes.

tissue (Fig. 1D). ELISA detection found that SHXY separately decreased the levels of the mouse serum inflammatory factors level of TNF-α and CRP (Fig. 1E). Although SHXY also displayed an effect on reducing the inflammatory factor level of IL-6, the results were not statistically significant.

A large number of M1 macrophages are M1macrophages located in DF wounds

Lasted single cell analysis revealed that the number of M1 macrophages was significantly increased in DF patient skin and blood (Theocharidis et al., 2022). In this part, we detected the M1 macrophages in DF mouse wounds. Immunofluorescence demonstrated that the M1 macrophage biomarkers iNOS and F4/80 were abundant in DF wounds (Fig. 2A). M1 macrophages showed a major proportion in immune cells in PBMCs and wound skin of DF patients through reanalyzing single-cell data in Georgios's study (Fig. 2B). M1 macrophage GO analysis showed that the biological process was mainly involved in the celluar response to oxygen levels and TNF-α, which were closely related to oxidative stress and the inflammatory response (Fig. 2C). These results suggested that M1 macrophages played the vital roles in the DF wound inflammatory response. Based on previous network pharmacology, we used 238 gene targets to intersect with the top 400 gene targets of M1 macrophages in PBMCs by Venn diagram. We found that genes of Nrf2 downstream protein HO-1 and HMGB1 were obtained from the intersection database (Fig. 2D).

SHXY upregulated the expression of AMPK, Nrf2, and HO-1 and decreased HMGB1 protein expression in DF mice

The roles of AMPK and Nrf2/HO-1 were also detected in the DF wound mouse models, and several related proteins were respectively measured by IHC (Fig. 3A) and Western blot (Fig. 3B). SHXY, metformin and fibroblast growth factor (FGF) were used to evaluate the expression of p-AMPK, Nrf2, and its downstream protein HO-1 in DF skin tissues on Day 7 and 14. Moreover, we found that serum HMGB1 expression was increased in DW group compared to the NW group, and administration of SHXY separately decreased its protein expression on Days 7 and 14 (Fig. 3C).

SHXY-containing serum activated p-AMPK, AMPK, and Nrf2/HO-1 in RAW 264.7 cells under a high glucose environment

In the previous part of the experiments, we observed that SHXY could effectively alleviate inflammation and activate AMPK and Nrf2/HO-1 pathways in two DF animal models with the down-regulation expression of HMGB1. Then, we established 30 mM high glucose (HG)-induced RAW 264.7 inflammation models *in vitro* to further explore the potential connection between these target proteins. We chose 10% 4. SHXY-containing serum used for *in vitro* experiments (Fig. 4A). SHXY-containing serum effectively decreased ROS, TNF-α and HMGB1 protein levels in the HG environment (Fig. 4A-F). Western blot detection also showed that SHXY increased the expression of p-AMPK, Nrf2 and HO-1 and decreased the expression of HMGB1 (Fig.4G).

SHXY-containing serum promoted Nrf2 translocation into the nucleus and inhibited HMGB1 extracelluar release in RAW 264.7 cells under high glucose environment

In this part, we used the Nrf2 inhibitor ML385 to treat RAW264.7 cells before SHXY drug serum during HG conditions. WB results showed that the inhibitory effect of SHXY on HMGB1 was weakened when Nrf2 expression was reduced by ML385 but had no influence on upregulating the expression of p-AMPK by SHXY. These results suggest that SHXY alleviated HG-induced *in vitro* inflammation by activating p-AMPK, Nrf2 and HO-1 and suppressing HMGB1, consistent with the *in vivo* experimental results, and Nrf2 could be a vital node in the AMPK/Nrf2/HMGB1 pathway (Fig. 5A). GSEA also suggested that Nrf2 existed significantly regulates on inflammatory response and defense response to bacteria in M1 macrophages (Fig.5B). Since Nrf2 is a nuclear transcription factor, we detected the influence of SHXY on the cellular location of Nrf2. Immunofluorescence showed that SHXY promoted Nrf2 expression in the cell nucleus compared with the HG group (Fig.5C). Moreover, SHXY inhibited HMGB1 extracelluar release. Western blot also showed that SHXY increased the expression of p-Nrf2 and Nrf2 in the cell nucleus (Fig.5D).

Table 2
. Information for quantitative analysis from UHPLC-Q-Orbitrap HRMS.

Number	No. Rt/ min	Molecular formula	Identification	Source	Peak area
1	23.95	C21H18O11	Baicalin	HQ	3,449,659,384
2	1.47	C6H8O7	Citric acid	SDH/ HZ/ BLG/ HQ/ HL/ HB/ LQ/ YZC/ QS/ GC/SZ	3,373,576,336
3	20.01	C20H17NO4	Berberine	HB/HL	3,058,116,717
4	27.20	C22H20O11	Wogonoside	HQ	2,558,985,366
5	1.10	C4H6O5	Malic Acid	SDH/ HZ/ BLG/ HQ/ HL/ HB/ LQ/ YZC/ QS/ GC/SZ	1,325,182,930
6	39.56	C15H10O5	Emodin	HZ/ SDH	1,220,921,832
7	0.94	C5H9NO2	l-Proline	SDH/ HZ/ BLG/ HQ/ HL/ HB/ LQ/ YZC/ QS/ GC/SZ	1,189,258,416
8	20.60	C21H22NO4	Palmatine	HB/HL	1,187,143,908
9	11.75	C20H23NO4	Phellodendrine	HB	1,177,760,465
10	13.83	$C_{17}H_{20}O_9$	Isomers of ferulic quinic acid	HB/ HL/ QS/ YZC	957,031,024

Identification of the components in SHXY

The accurate relative molecular weights of primary mass spectrometry were obtained from the information of excimer ions and charged ions provided by high-resolution mass spectrometry. The molecular formulas were fitted by Xcalibar 3.0 software and matched with CD (Compound discovery, 2.1) compound analysis and identification software and the self-built local library. The components obtained by high resolution Mass spectrometry were preliminarily speculated in the early stage, and then the relative retention time and the information of fragments and ions generated by high-energy collisions provided by the databases such as reference materials, references, Mass Bank and Chemical Book were further analysed accurately and qualitatively for the chemical components contained in SHXY prescription. The HPLC analysis identified 162 chemical components of SHXY, including 64 flavonoids, 18 organic acids, 16 alkaloids, 14 anthraquinones, 12 nucleosides, 12 amino acids, 4 saponins and 22 unknown ingredients (Fig. 6, Table 2 and Supplementary 4).

SHXY ameliorated inflammation in DF rats

Bacterial infection is a typical clinical feature of DF. Staphylococcus aureus (SA) is a major nosocomial and community-associated pathogen, that significantly contributes to wound infections as well(Shettigar and Murali, 2020). In this part, we also used STZ- and SA-induced infectious diabetic foot to assess the *in vivo* impact of SHXY (Fig. 7A). Compared with the STZ+SA group, SHXY significantly improved high glucose (HG) of DF rats (Fig. 7C), but had no effect on weight loss (Fig. 7D). The foot histological studies also showed that administration of SHYX for 1 month reduced inflammatory cell infiltration and deposition of ECM in tissue (Fig. 7B). SHXY also effectively decreased the levels of the rat serum inflammatory factors level of IL-6, TNF-α and CRP as determined by ELISA, which was consistent with the result in DF mice (Fig. 7E). The role of AMPK and the Nrf2/HO-1 pathway and several related proteins were respectively measured by immunohistochemistry (IHC) (Supplementary. 2A) and Western blot (Supplementary. 2B) in DF rat infection models. In present study, we observed that SHXY affected the expressions of p-AMPK and Nrf2 downstream protein HO-1 in DF foot tissues. At the same time, we found that HMGB1 expression was increased in the STZ+SA groups as compared to the untreated control, but the inhibition of SHXY on HMGB1 was insignificant. These results suggested that SHXY not only improved inflammatory microenvironment in DF mouse, but also had effect on DF rat.

Discussion

Diabetic foot (DF) is a serious complication of diabetes characterized by abnormal and chronic inflammation leading to foot infection and wound healing delay. Western medicine treatment focuses on single target therapy easily causing drug resistance and failed wound healing. Therefore, exploring the pathogenesis of DF and searching for new targets to combat DF are urgently needed, and possess great significance for clinical treatment. Traditional Chinese medicines possesses excellent therapeutic effects on immune diseases through regulating multiple target functions (Zhu et al., 2022). In this study, we first demonstrated that our classical hospital preparation SHXY possessed significant therapeutic effects on DF by alleviating abnormal inflammation in diabetic foot infection and diabetic wound healing, two mainly clinical feature of DF. Moreover, we also demonstrated that AMPK, Nrf2, HO-1 and HMGB1 could be the vital targets for SHXY treatment of DF. Finally, we identified the components in SHXY by UHPLC.

Foot-related disorders, including infection, ulceration, and gangrene, are a frequent indication for hospitalization of diabetic patients. Previous studies demonstrated that DF could escalate inflammation, including proinflammatory mediator (IL-6, TNF-α and CRP) formation, further damaging the foot skin barrier (Viswanathan et al., 2018). In accordance with these results, we found that SHXY decreased the protein levels of IL-6, TNF-α and CRP in serum from DF rats and a DF mouse model. In addition, we used histological studies to observe the number of infiltrating inflammatory cells in foot and skin tissues and found that SHXY can significantly suppress inflammatory cells infiltration and influence ECM formation. SHXY also emerged a prominent effect on promoting wound healing. *In vitro*, we used high glucose-treated macrophages to establish an inflammation model. SHXY drug serum also exhibited an anti-inflammatory effect through reducing the inflammatory protein levels *in vitro*. Based on above experiments SHXY was proved to have effect on improving abnormal inflammatory microenvironment in DF. Recent studies have demonstrated that macrophages exist as M1 and M2 types (Louiselle et al., 2021). The M1 type promotes inflammation effect and the M2 type has effect on anti-inflammatory effects. Single-cell analysis of DF patients suggested that the number of M1 macrophages was increased in blood and wound. Consistent with our study, we found a large population of M1 macrophages located in DF tissue. Many researches have proven high glucose easily promotes macrophage translation to the M1 type (Yu et al., 2019; Zhang et al., 2021). Whether the anti-inflammatory effect of SHXY occurs by regulating macrophage polarization should be further explored in our future work. Interestingly, we also observed that SHXY improved hyperglycaemia in a diabetic foot rat model, but this effect was not significant in a diabetic mouse model in our studies. This may be due to the SHXY administration time for rats being longer than that for mice.

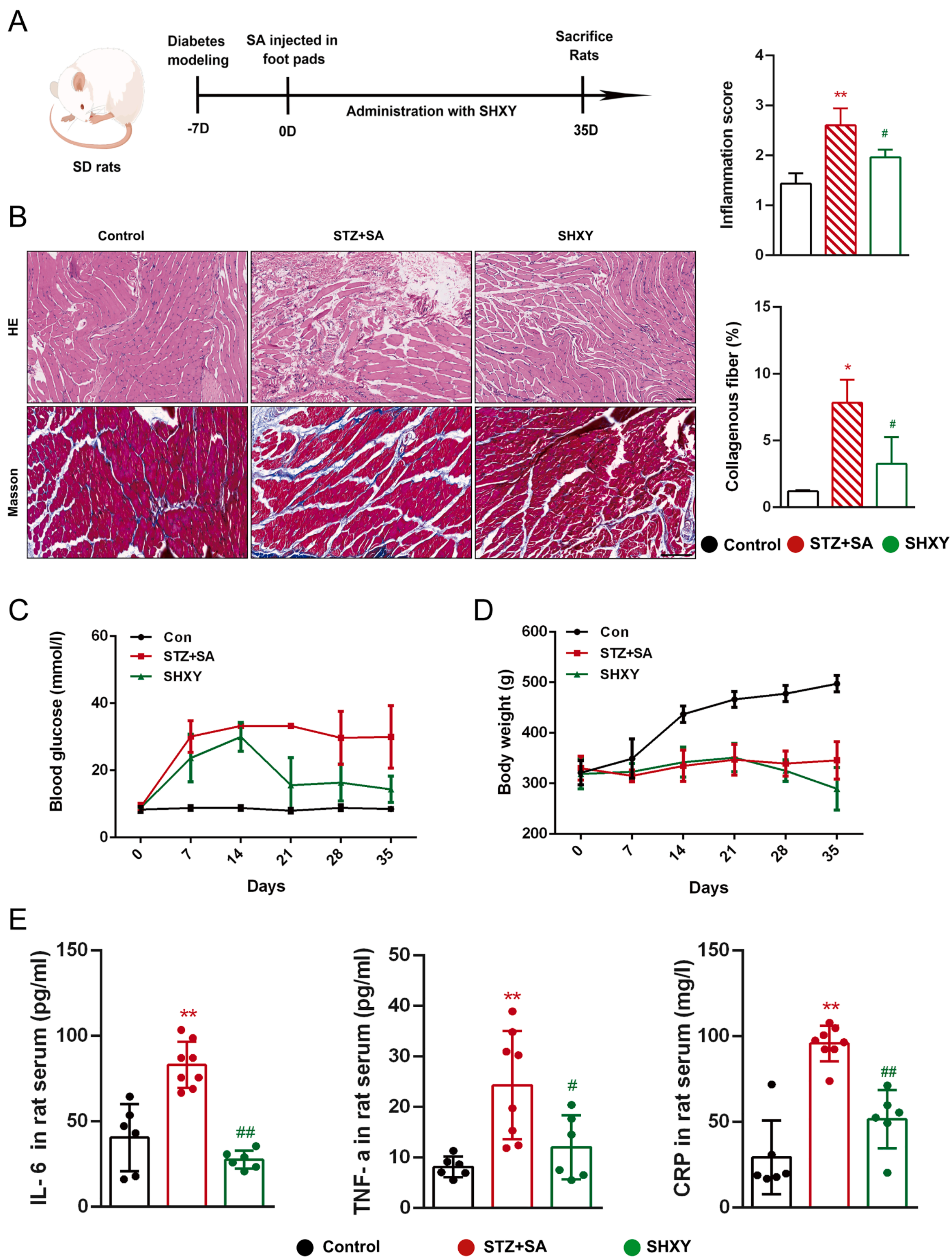

Fig. 7. Administration of SHXY reduced inflammation and improved ECM deposition in the DF rat infection model. (A) After the DF rat infection model established by STZ and SA. Rats were intragastric administration by SHXY (1.47 g/kg) for 35 days. All animals were euthanized on Day 36. (B) H&E and Masson staining of foot tissues from each group, respectively. Scale bars, 100 μm and 50 μm. (C) Blood glucose of rats in different groups, $n \geq 6$ per group. (D) Body weight of rats in different groups, $n \geq 6$ per group. (E) Inflammatory cytokine levels in serum, $n \geq 6$ per group. Values are shown as mean ± SD of 3 mice. * $p < 0.05$, ** $p < 0.01$ *vs* Con group and $^{\#}$ $p < 0.05$, $^{\#\#}$ $p < 0.01$ *vs* STZ+SA group.

In addition, we investigated three target proteins, AMPK, Nrf2 and HMGB1, that were involved in the anti-inflammatory process of SHXY in DF rat and DF mouse models. Many studies have proven that upregulating the protein expression of AMPK and Nrf2 can accelerate wound healing. Huangbai liniment accelerated diabetic wound healing by activating Nrf2 signalling (Zhang et al., 2020). The AMPK activator metformin promoted the M2 macrophage polarization to accelerate the wound healing (Qing et al., 2019). Moreover, many studies have

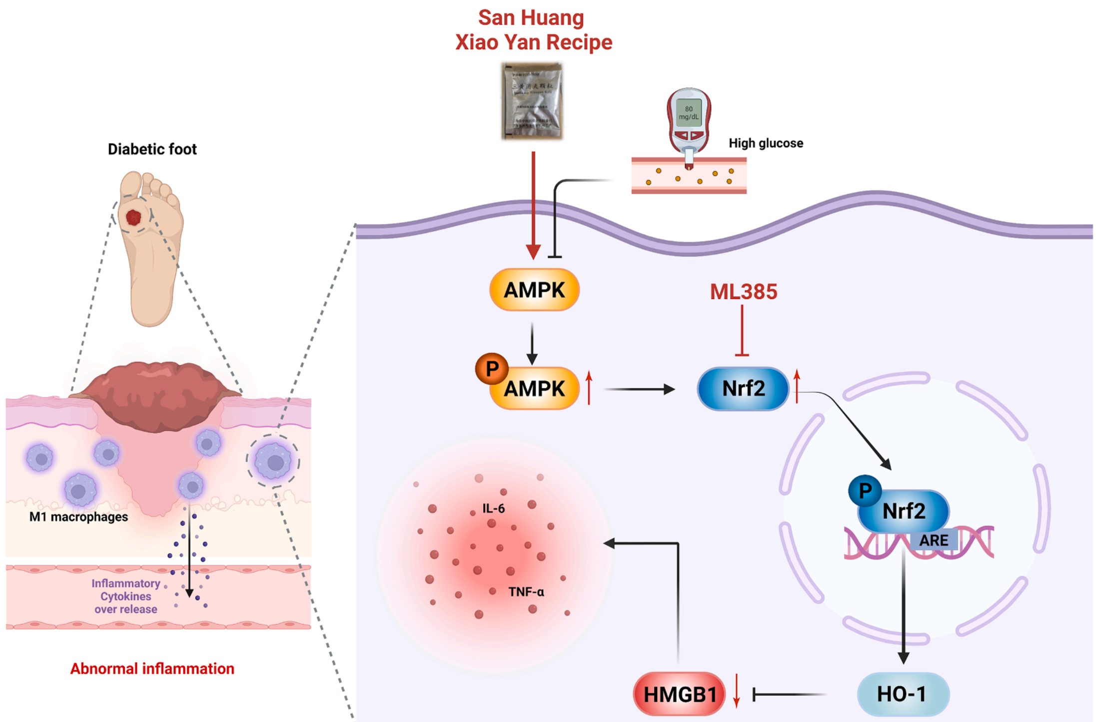

Fig. 8. SHXY ameliorates DF-associated inflammation via activating AMPK/Nrf2/HMGB1 pathway. SHXY activates Nrf2 by phosphorylating AMPK to further reduces inflammatory factors release via inhibiting HMGB1 expression in DF.

suggested that AMPK acts as an upstream regulatory factor for Nrf2 involving in many diseases. Activating the expression of AMPK in diabetic cardiomyopathy can upregulate the expression of Nrf2 by decreasing the level of ROS, thus accelerating the recovery of cardiac function (Li et al., 2018a). Overexpression of AMPK cannot improve the phenomenon of brain infarction and neurological deficits when Nrf2 expression is inhibited (Yu et al., 2020). AMPK/Nrf2 pathway activation improved SA-induced inflammation in acute lung injury (Wu et al., 2021). Nrf2 activator sulforaphane prevents ferroptosis and associated pathogenesis via AMPK-mediated Nrf2 activation (Wang et al., 2022). Phosphorylated AMPK could further activate phosphorylated Nrf2 at Ser40, promoting Nrf2 translation into the nucleus (Gallego-Selles et al., 2020). However, the potential relationship between AMPK and Nrf2 in DF is still a mystery. In present study, we found that SHXY and AMPK activator metformin upregulated the expression of AMPK and Nrf2 *in vivo* and *vitro*. We also found the expression of Ser 40 phosphorylated Nrf2 was increased in cell nucleus after SHXY treatment. Based previous literatures investigation and our experiences, we speculated that AMPK could be the upstream protein of Nrf2 for the mechanism of SHXY treating DF. In subsequent experiments, Ser40 of Nrf2 will be mutated to further verify whether this active site plays a key role in DF pathological mechanism. The elevated of the HMGB1 protein level in the serum of patients with diabetic foot (Hafez et al., 2018) and abnormal expression of HMGB1 could delay skin wound healing (Hoste et al., 2019). In our study, we also observed that SHXY markedly activated on AMPK and Nrf2 with the inhibiting effect on HMGB1. In sepsis and myocardial injury, Nrf2 can inhibit the activity of HMGB1 by activating HO-1 to further attenuate the pathological process of these two diseases (Kim et al., 2013; Wang et al., 2015). The Chinese herbal compound Dao-Chi powder could ameliorate pancreatitis-induced intestinal and cardiac injuries via regulating the Nrf2-HO-1-HMGB1 pathway in rats (Yao et al., 2022). These studies suggested that HMGB1 may be a downstream of Nrf2 in the process of DF. In present study, we used Nrf2 inhibitor ML385 to further explored the potential relationships between AMPK, Nrf2 and HMGB1 in DF. ML385 could relieve the inhibitory effect of SHXY on HMGB1 *in vitro* with no influence on AMPK. The above results indicate that the DF therapeutic effect of SHXY was mediated by the AMPK/Nrf2/HMGB1 pathway. At the same time, we observed that serum HMGB1 was more sensitive to SHXY in the DF pathological process than other inflammatory factors *in vivo*. Consistent with the *in vivo* results SHXY significantly inhibit HMGB1 extracellular release by elisa and immunofluorescence. HMGB1 as a key node, can promote inflammatory factor production by activating the TLR4/MYD88/NF-κb signalling pathway in many diseases (Liu et al., 2021b; Meng et al., 2018). Serum HMGB1 could be early diagnostic markers for DF superior to conventional inflammatory factors such as IL-6, TNF-α and CRP. Furthermore, a previous study showed HMGB1 could be a growth factor only existing in keratinocytes to promote wound healing (Straino et al., 2008). In our study, we found that HMGB1 did not only locate in epidermis but also in dermis by IHC. The role of HMGB1 in skin effector cells such as fibroblasts, keratinocytes and epithelial cells deserves further exploration in our next study.

In our research we also used ULPCs to identify specific ingredients in SHXY. We identified 162 chemical components. According to peak area, the top 10 ingredients were rhein, baicalin, citric acid, berbine, Wogonoside, malic acid, rheum emodin, l-proline, palmatine and phellodendrine. Rhein, baicalin, berberine, wogonoside, emodin and palmatine possess significant anti-inflammatory effects in many diseases through upregulating the protein expression of Nrf2. Baicalin alleviated oxidative stress and inflammation in diabetic nephropathy (DN) via the Nrf2 and MAPK pathways (Ma et al., 2021). Berberine reduced renal cell pyroptosis in golden hamsters with DN through the Nrf2-NLRP3-Caspase-1-GSDMD pathway (Ding et al., 2021). Emodin ameliorates renal damage and podocyte injury in a rat model of diabetic nephropathy via regulating AMPK/mTOR-mediated autophagy signalling pathway (Liu et al., 2021a). Palmatine promoted corneal wound healing by decreasing the expression levels of HMGB1 in diabetic keratopathy (Li et al., 2022). Whether these chemical monomers from SHXY also possess therapeutic effects on DF by activating the AMPK/Nrf2/HMGB1 pathway deserves deep exploration in our future study.

Conclusions

In conclusion, SHXY improved the inflammation and poor wound healing via activating the AMPK/Nrf2/HMGB1 pathway, thus alleviating STZ-induced diabetic foot (Fig. 8).

CRediT authorship contribution statement

Zhihui Zhang: Writing – original draft, Investigation, Formal analysis. **Yihan Zheng, Nan Chen, Chenqin Xu and Jie Deng:** Investigation, Formal analysis. **Xia Feng, Tongkai Cai and Yicheng Xu:** Methodology. **Jian Chen, Chao Ma and Wei Liu:** Network pharmacology, Composition analysis of traditional Chinese medicine and single-cell re-analysis guidance. **Song Wang:** Drug resources. **Yemin Cao:** Resources. **Chenglin Jia:** Supervision. **Guangbo Ge:** Supervision, Project administration. **Yongbing Cao:** Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

All authors declare no conflict of interest.

Acknowledgments

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.phymed.2023.154931.

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